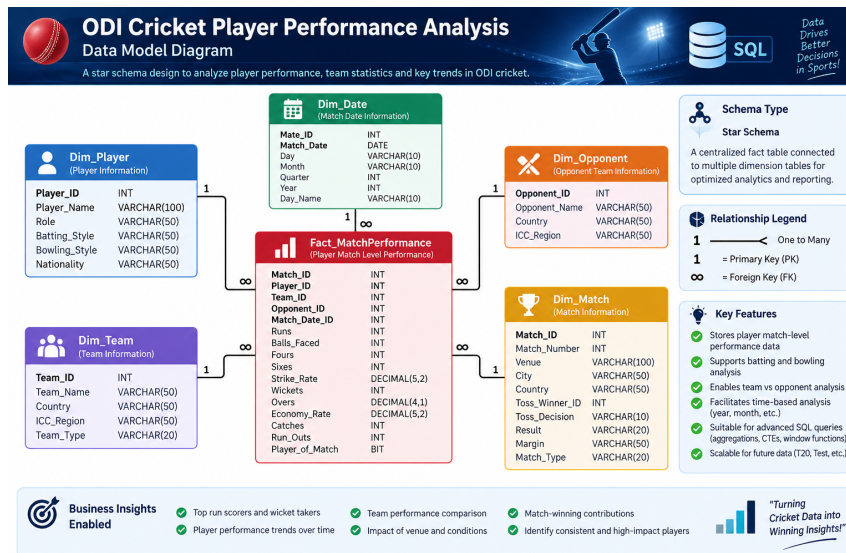


ODI Cricket Player Performance Analysis

An End-to-End SQL Analytics Project



Project Overview

This project analyzes One Day International (ODI) cricket player performance using Microsoft SQL Server. It transforms raw cricket statistics into structured insights for identifying top performers, performance patterns and key trends in the ODI format.

Repository-ready documentation

Database • SQL queries • CSV results • Data model • Analytical framework • Business insights

1. ■ Executive Summary

The ODI Cricket Player Performance Analysis project is an end-to-end SQL analytics project focused on turning player-level ODI statistics into understandable performance insights. The analytical workflow is organized into four phases: KPI Analysis, Univariate Analysis, Bivariate Analysis and Multivariate Analysis.

The project is built around a Microsoft SQL Server database and the **cricsheet** table. The supplied project material describes a dataset containing **1,540 players** and player-performance fields covering batting, bowling, matches, wins/losses and Player of the Match awards.

The objective is not simply to list statistics, but to structure the data so that a GitHub visitor can understand how raw records move through the database and SQL analysis layers to produce meaningful cricket-performance conclusions.

2. ■ Project Objectives

- Create a structured SQL Server database for ODI player-performance data.
- Analyze batting, bowling, match participation and player-impact statistics.
- Identify top run scorers, strike-rate leaders and leading wicket takers.
- Compare performance across teams and player roles.
- Explore relationships between experience, performance and match success.
- Use SQL techniques such as aggregations, CTEs, subqueries, CASE expressions and window functions.
- Generate reusable CSV query results for reporting and visualization.

3. ■ Dataset Overview

Item	Details
Dataset	ODI Cricket Player Performance Dataset
Records	1,540 players
Database	Microsoft SQL Server
Main table	cricsheet
Analysis style	Player-performance analytics
Primary output	SQL query results and business-style insights

The supplied project overview identifies the main dataset fields as Player Name, Role, Team, Total Runs, Strike Rate, Total Wickets, Runs Conceded, Matches Played, Wins, Losses and Player of the Match Awards, along with additional performance fields.

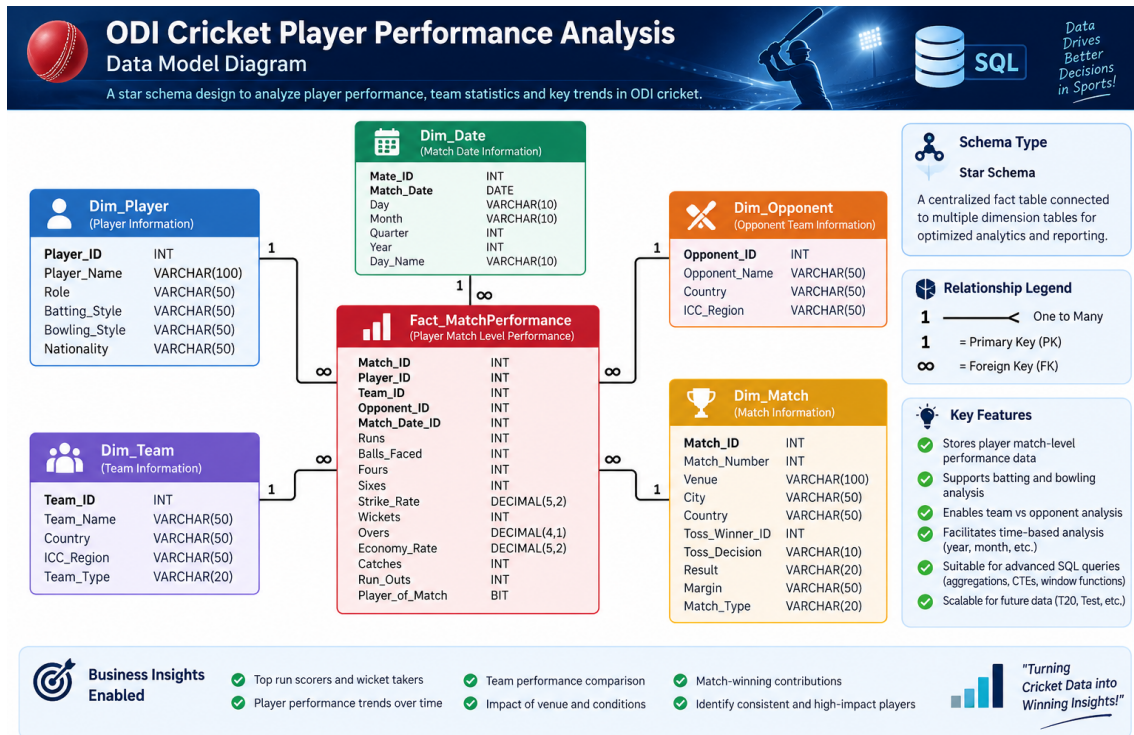
4. ■■ Data Model

The current project is centered on one analytical table rather than a multi-table transactional schema. This is appropriate for the supplied player-level dataset because the analysis is performed directly across player attributes and performance measures.

Column	Purpose / Interpretation
player_name	Unique player-level identifier used for ranking and grouping.
role	Player role classification used for role comparisons.

Column	Purpose / Interpretation
team	Team associated with the player.
total_runs	Total ODI runs used for batting-performance analysis.
strike_rate	Batting strike-rate metric.
total_balls_faced	Batting workload / experience measure.
total_wickets_taken	Wickets taken for bowling-performance analysis.
total_runs_conceded	Runs conceded while bowling.
total_overs_bowled	Bowling workload.
total_matches_played	Overall ODI participation.
matches_played_as_batter	Matches in batting role.
matches_played_as_bowler	Matches in bowling role.
matches_won	Matches associated with wins.
matches_lost	Matches associated with losses.
player_of_match_awards	Player-of-the-Match impact indicator.
average	Player batting-average measure where available.

5. ■ Data Model Diagram



Model interpretation: The *cricsheet* table acts as the central analytical fact table. Player, role and team are categorical dimensions, while runs, strike rate, wickets, matches and awards are performance measures. SQL aggregations and analytical functions operate across these fields to create the four analysis phases.

6. ■■ Tools & Technologies

- **Microsoft SQL Server** — database platform.
- **SQL Server Management Studio (SSMS)** — database and query execution environment.
- **SQL / T-SQL** — analytical language.
- **CTEs** — readable multi-step analytical queries.
- **Aggregations** — SUM, AVG, COUNT and grouped performance measures.
- **Window Functions** — ranking, comparison and advanced player analysis.
- **Subqueries** — nested analytical logic.
- **CASE WHEN** — performance classification and conditional analysis.

7. ■ Analytical Framework

Phase	Focus	Purpose
1	KPI Analysis	Establish high-level performance indicators.
2	Univariate Analysis	Study individual player-performance variables.
3	Bivariate Analysis	Study relationships between two variables.
4	Multivariate Analysis	Combine multiple variables for deeper insights.

8. ■ Phase 1 — KPI Analysis

- Total unique players
- Total matches played
- Total runs scored
- Total wickets taken
- Average player runs
- Average strike rate
- Average wickets
- Total wins and losses
- Player of the Match awards
- Top run scorer
- Top wicket taker
- Highest strike rate
- Most experienced players
- Most successful players
- Overall average player performance

9. ■ Phase 2 — Univariate Analysis

- Player role distribution
- Team-wise player distribution
- Runs distribution
- Strike-rate distribution
- Wickets distribution
- Matches played distribution
- Wins distribution
- Losses distribution
- Player of the Match distribution
- Batting participation
- Bowling participation
- Top run scorers
- Top wicket takers
- Top strike-rate performers
- Bowling workload / participation analysis

10. ■ Phase 3 — Bivariate Analysis

- Runs vs Strike Rate
- Runs vs Balls Faced
- Wickets vs Runs Conceded
- Wickets vs Overs Bowled
- Matches vs Runs
- Matches vs Wickets
- Wins vs Runs
- Wins vs Wickets
- Player of the Match vs Runs
- Player of the Match vs Wickets
- Experience vs Strike Rate
- Experience vs Average
- Team vs Average Performance
- Role vs Performance
- Strike Rate vs Player Average

11. ■ Phase 4 — Multivariate Analysis

- Overall player-performance scorecards.
- Top players by combined batting and bowling indicators.
- Team and role performance comparisons.
- Experience, success and performance relationships.
- Match-winning contribution analysis.
- Player-of-the-Match impact analysis.
- Combined batting workload and output analysis.
- Combined bowling workload and output analysis.
- Cross-comparison of runs, wickets and match success.
- Elite-player identification using multiple measures.

12. ■ Example SQL Analysis

```
SELECT TOP 10
player_name,
total_runs,
strike_rate
FROM cricsheet
ORDER BY total_runs DESC;
```

This example ranks players by total ODI runs and returns the top 10 performers. Similar SQL patterns are extended throughout the four analytical phases.

13. ■ Key Insights Framework

- Identify top run scorers and strike-rate leaders.
- Identify leading wicket takers and bowling-performance leaders.
- Compare player performance across teams and roles.
- Explore the relationship between experience and success.
- Measure match-winning contributions.
- Highlight Player-of-the-Match impact.

14. ■ Query Results & GitHub Outputs

For portfolio presentation, each analytical query can be exported as a normal CSV file. The recommended repository keeps results grouped by phase so visitors can quickly inspect the output of each analysis stage.

Folder	Contents
Database/	SQL Server database creation and data-load script.
Queries/	Four-phase SQL analysis script.
Results/Phase_1_KPI/	KPI query-result CSV files.
Results/Phase_2_Univariate/	Univariate query-result CSV files.
Results/Phase_3_Bivariate/	Bivariate query-result CSV files.
Results/Phase_4_Multivariate/	Multivariate query-result CSV files.
Dashboard/	Project dashboard / visualization.
Reports/	This project documentation PDF.

15. ■ Recommended GitHub Repository Structure

ODI-Cricket-Player-Performance-Analysis/

■

■ ■ ■ ■ ■ Database/

■ ■ ■ ■ ■ ODI_Cricket_Performance_Database.sql

■

■ ■ ■ ■ ■ Queries/

■ ■ ■ ■ ■ SQL_QUERY.sql

■

■ ■ ■ ■ ■ Dataset/

■ ■ ■ ■ ■ Cricsheet.csv

■

■ ■ ■ ■ ■ Results/

■ ■ ■ ■ ■ Phase_1_KPI/

■ ■ ■ ■ ■ Phase_2_Univariate/

■ ■ ■ ■ ■ Phase_3_Bivariate/

■ ■ ■ ■ ■ Phase_4_Multivariate/

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■ ■ ■ ■ ■ Dashboard/

■ ■ ■ ■ ■ Dashboard.png

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■ ■ ■ ■ ■ Reports/

■ ■ ■ ■ ■ ODI_Cricket_Player_Performance_SQL_Project_Report.pdf

■

■ ■ ■ ■ ■ README.md

16. ■■ Project Workflow

Step	Activity	Output
1	Collect ODI player dataset	Raw player-performance data
2	Create SQL Server database	ODI performance database
3	Create / populate cricsheet	Structured analytical table
4	Validate records and columns	Ready-to-query dataset
5	Run KPI analysis	High-level metrics
6	Run univariate analysis	Individual-variable insights
7	Run bivariate analysis	Relationship insights
8	Run multivariate analysis	Combined performance insights
9	Export CSV results	Reusable analysis outputs
10	Document and visualize	GitHub-ready portfolio project

17. ■ Testing & Validation

- Confirm the database is created successfully.
- Confirm the cricsheet table exists with the required columns.
- Confirm the expected 1,540 player records are loaded.
- Check numeric columns for valid numeric values.
- Run sample SELECT queries before executing advanced analysis.
- Validate ranking queries using descending performance measures.
- Check aggregate results against the source dataset where appropriate.
- Verify exported CSV files contain headers and readable results.

18. ■■ Limitations

- The supplied project material describes a player-level analytical dataset rather than ball-by-ball event data.
- The current model is centered on a single cricsheet table.
- The dataset does not provide every contextual factor that can affect cricket performance.
- Predictive modeling is outside the core SQL scope.
- The project should not be interpreted as a causal model of match outcomes.

19. ■ Future Enhancements

- Add ball-by-ball ODI data for richer tactical analysis.
- Create separate Player, Team, Match and Performance dimension/fact tables.
- Build a Power BI interactive dashboard.
- Add career-period and year-over-year comparisons.
- Add venue, opposition and match-condition analysis where data is available.
- Introduce predictive models for player performance.

- Automate CSV export and scheduled reporting.
- Add player comparison pages for portfolio presentation.

20. ■ Conclusion

The ODI Cricket Player Performance Analysis project demonstrates how SQL can be used as a complete analytical workflow rather than only as a data-retrieval language. Starting with structured player records, the project builds from baseline KPIs to single-variable exploration, relationship analysis and multi-variable performance evaluation.

The resulting database, SQL scripts, CSV outputs, data model and documentation provide a clear portfolio story: **Data** → **Database** → **SQL Analysis** → **Results** → **Insights**. This makes the repository understandable to recruiters, reviewers, classmates and anyone interested in practical SQL analytics.

21. ■ Key Definitions

Term	Meaning
KPI	Key Performance Indicator used to summarize performance.
Univariate	Analysis of one variable or dimension at a time.
Bivariate	Analysis of the relationship between two variables.
Multivariate	Analysis involving several variables together.
Strike Rate	Batting scoring-rate measure.
Player of the Match	Award indicator used to represent match impact.
CTE	Common Table Expression used to structure multi-step SQL logic.
Window Function	SQL analytical function used for ranking, comparisons and calculations across related rows.

■ ODI Cricket Player Performance Analysis
Data • SQL • Insights • Better Understanding of Cricket Performance